

A Sufficient Lower Bound for Sampling of Multiple Correlated Stochastic Signals

Lin Jin^a, Hang Sheng^a, Hui Feng^{a,b,*}, and Bo Hu^{a,b}

^a College of Future Information Technology, Fudan University, Shanghai 200433, China

^b State Key Laboratory of Integrated Chips and Systems, Fudan University, Shanghai 200433, China

Abstract—Multiple stochastic signals received in array or MIMO systems may have inherent statistical correlations, yet conventional sampling on each channel independently result in data redundancy. To leverage such correlation for efficient sampling, we model correlated signals as a linear combination of a smaller set of uncorrelated, wide-sense stationary (WSS) latent sources. We establish a sufficient lower bound on the total sampling density for zero mean-square error reconstruction. We then develop a constructive multi-band sampling scheme to achieve this theoretical bound. The proposed method operates via spectral partitioning, followed by spatio-temporal sampling and interpolation. Experiments on synthetic and real datasets confirm that our scheme achieves near-lossless reconstruction precisely at the theoretical sampling density.

Index Terms—Correlated signals, wide-sense stationary signal, sampling theory

I. INTRODUCTION

Multi-channel stochastic signals, which we term multiple stochastic signals in this paper, arise across many modern signal-processing applications and often serve as the primary information-bearing quantities in various complex systems. Notably, these channels are frequently not independent but instead inherently exhibit statistical correlation. For instance, in sensor arrays such as radar and sonar, common physical sources induce coherent signatures across sensors [1]; in Multiple-Input Multiple-Output (MIMO) wireless receivers, spatial and propagation effects induce inter-channel correlation [2]; and in multi-microphone audio captures, a single acoustic source creates correlated recordings across sensors [3]. The structure embedded in these correlations contains system-level information unavailable from any single channel. Leveraging such correlation therefore becomes critical to increase signal processing efficiency.

The proliferation of multiple systems leads to the generation of massive datasets, imposing heavy burdens on communication, storage, and computational resources [4]. Efficiently converting these continuous-time signals into discrete observations through sampling is therefore a critical necessity for mitigating resource costs. In multiple scenarios, naively sampling each channel independently ignores the information reuse potential offered by inter-channel correlation, leading to redundant data acquisition. Consequently, developing sampling theories that explicitly exploit correlation to minimize sampling cost is of paramount importance.

For single-channel signals, the classical Shannon-Nyquist sampling theorem is a cornerstone for deterministic signals. It is less well known that the theorem can also extend to wide-sense stationary (WSS) stochastic signals with bandlimited power spectral density (PSD) [5], yielding a sufficient sampling rate condition. Existing studies have investigated sampling of single stochastic signals [6], [7], but they do not address the multiple signals.

As to multiple stochastic signals, research has explored correlation from various perspectives. Shlezinger et al. [8] proposed a distributed sampling and joint reconstruction framework, demonstrating the

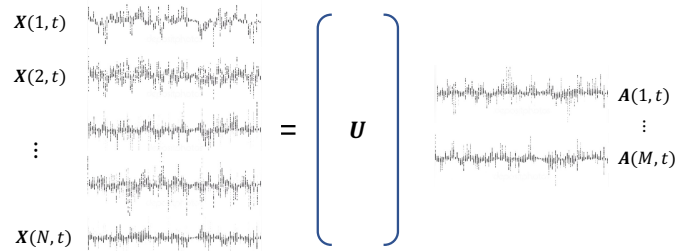


Fig. 1: Correlated signal model.

benefit of leveraging correlations; nevertheless, a theoretical analysis remains lacking. Theories rooted in compressed sensing have partially addressed rate reduction, yet their guarantees rely on strong sparsity assumptions, limiting their applicability to more general signals [9]–[11]. Based on a latent source model, Ahmed and Romberg [12], [13] proposed a random mixing architecture with corresponding sampling rate bound. However, their approach requires a dedicated analog front-end to physically mix signals before sampling and targets deterministic waveform recovery rather than stochastic sampling guarantees. In summary, there is a lack of comprehensive theoretical guarantees on the sampling lower bound for multiple correlated stochastic signals, and practical schemes that attain this bound are still less explored.

We model the correlated signals as linear combinations of low-dimensional latent sources and establish a sufficient lower bound on the total sampling density required for reconstruction with zero mean squared error (MSE). The derived bound is a sufficient achievable sampling-density condition under the proposed model, in the sense that any total sampling density no smaller than this bound is guaranteed by our construction to admit zero-MSE reconstruction. This result reveals that the requisite sampling budget is determined solely by the latent source bandwidth, effectively decoupling the total information rate from the number of correlated signals. Building upon this bound, we then propose a constructive multi-band sampling scheme that achieves this sufficient rate via power spectrum partitioning.

The main contributions of this paper are: (i) Under the latent source generation model, we prove that the sufficient lower bound on total sampling density for reconstruction with zero MSE equals the ratio of the latent signals' total spectral bandwidth to the number of correlated signals; (ii) We propose a constructive multi-band sampling scheme to achieve the derived lower bound while ensuring reconstruction in the mean-square sense; (iii) We validate the effectiveness of the proposed scheme through experiments on both synthetic and real datasets, demonstrating that the scheme realizes near-lossless reconstruction at the theoretical sampling density.

* Corresponding author.

II. SIGNAL MODEL

Let $\mathbf{X} = [X(1, t), X(2, t), \dots, X(N, t)]^T, \forall t \in \mathbb{R}$ denote N -channel correlated stochastic signals, where each channel $X(n, \cdot), n = 1, \dots, N$ is a continuous-time signal. We establish a correlated signal model inspired by [12], in which correlated signals can be represented as linear combinations of a smaller number of latent signals. Specifically, we assume that each channel $X(n, \cdot)$ is a linear mixture of M mutually uncorrelated signals $\mathbf{A} = [A(1, t), \dots, A(M, t)]^T, \forall t \in \mathbb{R}$:

$$\mathbf{X} = \mathbf{U}\mathbf{A}. \quad (1)$$

Each channel of the latent signals $\mathbf{A}$ is a WSS continuous-time signal and has finite power spectral bandwidth. The mixing matrix $\mathbf{U} \in \mathbb{R}^{N \times M}$ combines the M latent signals. The signal model is illustrated in Fig. 1.

The generative model aligns with standard physical mixing formulations in array signal processing. In this context, $\mathbf{X}$ denotes the signals received by the sensor array, and $\mathbf{U}$ represents the steering matrix, where each column corresponds to a steering vector associated with a specific Direction of Arrival (DoA). Consequently, the latent signals $\mathbf{A}$ correspond to the uncorrelated source waveforms impinging on the array from these distinct spatial locations. Similarly, in MIMO wireless communications, $\mathbf{U}$ serves as the channel propagation matrix for spatial multiplexing, mixing independent data streams $\mathbf{A}$ into the received signals. In both scenarios, the observed inter-channel correlation is physically induced by the linear mixture of these underlying low-dimensional sources, as defined in (1).

Denote the correlation function of $\mathbf{X}$ by

$$\mathbf{R}_X(i, j, \tau) = \mathbb{E} [X(i, t)X^H(j, t - \tau)], \quad (2)$$

where it reduces to the autocorrelation of a single channel when $i = j$, and to the cross-correlation between two channels when $i \neq j$. If there exist indices $i \neq j$ and lags τ such that $\mathbf{R}_X(i, j, \tau) \neq 0$, then the signals $X(i, \cdot)$ and $X(j, \cdot)$ exhibit inter-channel statistical correlation. In this case, the multiple stochastic signals $\mathbf{X}$ is referred to as correlated. Correspondingly, we define the latent stochastic signals $\mathbf{A}$ as uncorrelated if the cross-correlation function between any two channels is identically zero:

$$\mathbf{R}_A(i, j, \tau) \equiv 0, \quad \forall i \neq j, \forall \tau \in \mathbb{R}. \quad (3)$$

By the Wiener-Khinchin theorem, the auto-(or cross-) power spectral density of $\mathbf{X}$ is the Fourier transform of its auto-(or cross-) correlation function [14]: $\mathbf{S}_X(i, j, f) = \mathcal{F}\{\mathbf{R}_X(i, j, \tau)\}$. For brevity, we simplify the notation by retaining only the autocorrelation of $\mathbf{A}$, denote as $\mathbf{R}_A(m, \tau)$ for $m = 1, \dots, M, \forall \tau \in \mathbb{R}$, with the corresponding power spectral densities denoted by $\mathbf{S}_A(m, f)$.

Assume the power spectrum supports of $\mathbf{A}$ are known. For each channel m , let the spectral support set $\mathcal{B}_m \subset \mathbb{R}$ be Lebesgue-measurable with finite measure, i.e. $\mathbf{S}_A(m, f) = 0, \forall f \notin \mathcal{B}_m$. Denote by $\mu(\cdot)$ the Lebesgue measure and define the total power spectrum bandwidth of $\mathbf{A}$ as

$$B := \sum_{m=1}^M \mu(\mathcal{B}_m), \quad (4)$$

with $B < +\infty$.

For the n -th channel signal $X(n, \cdot)$, its temporal sampling set is denoted as $\mathcal{S}_n = \{t_{n,s} : t_{n,s} \in \mathbb{R}\}$. The overall sampling set for multiple signals $\mathbf{X}$ is

$$\mathcal{S} = \{(n_s, t_{n,s}) : n_s \in \{1, \dots, N\}, t_{n,s} \in \mathbb{R}\}. \quad (5)$$

We define its spatial projection as $\mathcal{S}_C = \{n_s : (n_s, t_{n,s}) \in \mathcal{S}\}$ and the temporal projection as $\mathcal{S}_T = \bigcup_{n_s \in \mathcal{S}_C} \mathcal{S}_n = \{t_{n,s} : (n_s, t_{n,s}) \in \mathcal{S}\}$.

To quantify the overall sampling cost of the N -channel signals $\mathbf{X}$, particularly for non-uniform sampling schemes, the total average sampling density is defined in [15] as

$$D(\mathcal{S}) := \liminf_{t \rightarrow \infty} \frac{|\mathcal{S} \cap \{\{1, \dots, N\} \times [-t, t]\}|}{2tN}, \quad (6)$$

where $|\cdot|$ denotes set cardinality.

III. SAMPLING THEOREM

Based on the signal model in Sec. II, we present a sampling theorem for multiple correlated stochastic signals that operates directly on the N channels, without any pre-sampling analog mixing stages. The key principle is that the information rate is governed by the spectral support of the M underlying, uncorrelated latent signals rather than the correlated signal channel count N . The proposed theorem effectively decouples the acquisition cost from the number of channels. Sec. III-A states the main result and sketches its proof; Sec. III-B then gives a constructive multi-band scheme to achieve the bound.

A. Main Result

Given the generative signal model (1), we now first specify the conditions under which $\mathbf{X}$ is guaranteed to be correlated.

Proposition 1. *Let the signals $\mathbf{X}$ be generated by the model in (1), $\mathbf{X}$ is guaranteed to be correlated if the following conditions hold: (i) The linear transformation is dimensionality-increasing, i.e., $N > M$; (ii) Each $A(m, \cdot)$ has positive power, i.e., $\mathbb{E}[|A(m, \cdot)|^2] > 0, \forall m \in \{1, \dots, M\}$; (iii) Each row of the mixing matrix $\mathbf{U}$ is a nonzero vector.*

Proof. A formal proof is provided in the extended version on arXiv [16]. $\square$

Theorem 1. *Let $\mathbf{A}$ be multiple uncorrelated stochastic signals with total spectral bandwidth $B < +\infty$, as defined in (4). Let $\mathbf{U} \in \mathbb{R}^{N \times M}$ be a mixing matrix with $\text{rank}(\mathbf{U}) = M$ that generates correlated signals $\mathbf{X}$ according to (1). Then, there exists a sampling set $\mathcal{S}$ satisfies*

$$D(\mathcal{S}) \geq \frac{B}{N} \quad (7)$$

such that a reconstruction $\hat{\mathbf{X}}$ of the signals $\mathbf{X}$ can be obtained, satisfying the zero MSE condition: $\mathbb{E}[\|\mathbf{X} - \hat{\mathbf{X}}\|^2] = 0$.

Proof. Since the total spectral bandwidth of $\mathbf{A}$ is B , the total sampling rate requires to reconstruct the multiple uncorrelated signals $\mathbf{A}$ is B . We can derive the sampling rates and temporal sampling sets from each subband of $\mathbf{A}$. The sampling set for the N -channel signals $\mathbf{X}$ after mixing is established by using the same temporal sampling set and obtaining the corresponding spatial sampling set via the linear transform $\mathbf{U}$. Thus $\mathcal{S}$ is derived, with total sampling rate B . Therefore, the total sampling density of $\mathbf{X}$ satisfies $D(\mathcal{S}) \geq \frac{B}{N}$. A detailed proof is given in the extended version on arXiv [16]. $\square$

The derived lower bound $D(\mathcal{S}) \geq \frac{B}{N}$ implies that the total sampling rate across all N channels is $N \cdot D(\mathcal{S}) = B$. In contrast to independent Nyquist sampling, where the total rate scales linearly with N , our result demonstrates that a sufficient acquisition rate is invariant to the channel count N , depending only on the total bandwidth B of the latent sources.

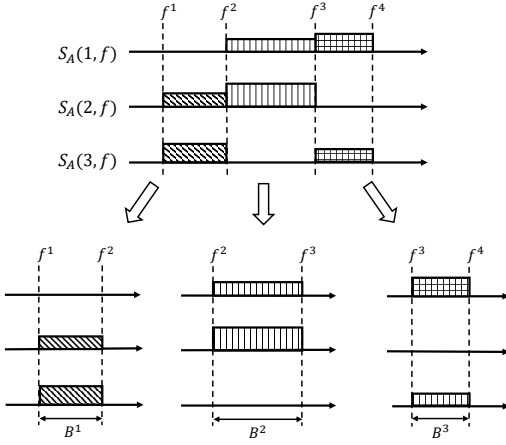


Fig. 2: Example of subband division.

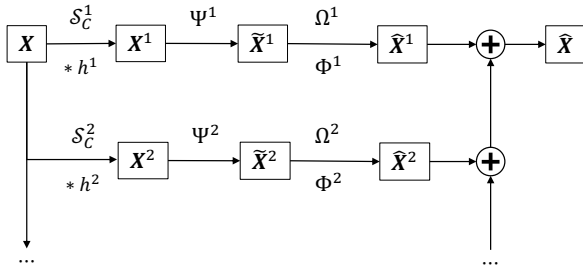


Fig. 3: Flow chart of the multi-band sampling scheme.

B. Multi-band Sampling Scheme

In this subsection, we construct a multi-band sampling scheme that samples $\mathbf{X}$ at the derived sufficient total sampling density $\frac{B}{N}$ and achieves zero MSE reconstruction. The scheme adapts the multi-band partitioning principle, previously explored for graph signals in [17], to the context of multiple stochastic signals.

We begin by partitioning $\mathbf{S}_A(\cdot, f)$ into L subbands, each corresponding to a frequency interval $[f^l, f^{l+1}]$. The boundaries of the frequency intervals are chosen at frequencies where the PSD of any latent source either starts or vanishes, which ensures that, within each subband, the PSD of a given source is either identically zero or nonzero. Fig. 2 provides an illustrative example of the subband division.

Sampling and reconstruction are performed independently within each subband. For subband $l \in \{1, \dots, L\}$, let Γ^l denote the index set of latent sources with nonzero PSD. The signal components corresponding to the l -th subband are extracted by convolving the correlated signals $\mathbf{X}$ with a band-pass filter, $h^l(t)$. The frequency response of this filter, $H^l(f)$, is equal to unity over $[f^l, f^{l+1}]$ and zero elsewhere. This operation yields the following relation for the subband signals: $\mathbf{X} * h^l = \mathbf{U}(\cdot, \Gamma^l) (\mathbf{A}(\Gamma^l, \cdot) * h^l)$.

Select $|\Gamma^l|$ linearly independent rows from $\mathbf{U}(\cdot, \Gamma^l)$, where Ξ^l is the index set. The spatially sampled signals are therefore $\mathbf{X}^l = \mathbf{U}(\Xi^l, \Gamma^l) (\mathbf{A}(\Gamma^l, \cdot) * h^l)$. For notational convenience set $\mathbf{U}^l := \mathbf{U}(\cdot, \Gamma^l)$ and $\mathbf{U}_s^l = \mathbf{U}(\Xi^l, \Gamma^l)$. The corresponding spatial sampling set is $\mathcal{S}_C^l = \{n_{s_1}, n_{s_2}, \dots, n_{s_{|\Gamma^l|}}\}$.

$\mathbf{X}^l(n, \cdot)$ for $n \in \mathcal{S}_C^l$ is sampled in time at the rate $f^{l,s} = f^{l+1} - f^l = B^l$. Denote the temporal sampling set by $\mathcal{S}_T^l = \{t_{n,s_1}, t_{n,s_2}, \dots\}$ where the asymptotic sampling density of $\mathcal{S}_T^l$ is

B^l , we have the multi-band sampled signals at the l -th subband:

$$\tilde{\mathbf{X}}^l(\mathcal{S}_C^l, \mathcal{S}_T^l) = \Psi^l \mathbf{X}^l, \quad (8)$$

where Ψ^l denotes the temporal sampling operator associated with $\mathcal{S}_T^l$.

In each subband, the sampled signals $\tilde{\mathbf{X}}^l(\mathcal{S}_C^l, \mathcal{S}_T^l)$ can be temporally interpolated by $\Omega^l = \text{sinc}(B^l t^l) e^{j2\pi f^{l,c} t^l}$ to obtain $\hat{\mathbf{X}}^l(\mathcal{S}_C^l, \cdot)$, where $f^{l,c} = \frac{f^l + f^{l+1}}{2}$ and $t^l = t - t_{n,s_i}$, $t_{n,s_i} \in \mathcal{S}_T^l$. To lift the reconstruction to all spatial channels, we apply the spatial interpolation operator

$$\Phi^l = \mathbf{U}^l (\mathbf{U}_s^l{}^H \mathbf{U}_s^l)^{-1} \mathbf{U}_s^l{}^H, \quad (9)$$

so that the full subband reconstruction is $\hat{\mathbf{X}}^l = \Phi^l \hat{\mathbf{X}}^l(\mathcal{S}_C^l, \cdot)$, which is valid because $\text{rank}(\mathbf{U}_s^l) = |\Gamma^l|$.

Finally, summing the reconstructions over all subbands produces the complete reconstruction

$$\hat{\mathbf{X}} = \sum_{l=1}^L \hat{\mathbf{X}}^l, \quad (10)$$

and this scheme attains zero MSE, with $\mathbb{E} \left[\|\mathbf{X} - \hat{\mathbf{X}}\|^2 \right] = 0$.

The process of the proposed scheme is shown in Fig. 3.

IV. EXPERIMENTS

In this section, we present experiments on synthetic and real data to validate the sampling theory developed in Sec. III and demonstrate the effectiveness of the proposed multi-band sampling scheme.

A. Results on synthetic dataset

To synthesize correlated signals, we first generate M -channel latent signals $\mathbf{A}(m, t)$, $m = 1, \dots, M$; $t = 1, \dots, T$, each a finite-length discrete-time random sequence with length T .

The statistical properties of $\mathbf{A}$ are defined by prescribing a PSD for each channel. $\mathbf{S}_A(m, \cdot)$ is procedurally generated as a symmetric, piece-wise constant function composed of randomly placed rectangular blocks:

$$\mathbf{S}_A(m, k) = \sum_{i=1}^{K_m} c_{m,i} \cdot \mathbb{I}(k \in \mathcal{F}_{m,i}), \quad (11)$$

where k is the discrete frequency index, K_m is the number of blocks for channel m , $c_{m,i}$ are their power levels, and $\mathcal{F}_{m,i}$ are disjoint frequency intervals.

Each signal $\mathbf{A}(m, t)$ is then synthesized as a discrete, finite-length and real-valued sample function of a WSS process with the prescribed PSD by summing sinusoidal components with amplitudes derived from the PSD and the randomized phases. For an even signal length T , the resulting time-domain expression is:

$$\begin{aligned} A(m, t) = & \frac{1}{\sqrt{T}} \left(\sqrt{\mathbf{S}_A(m, 0)} + (-1)^t \sqrt{\mathbf{S}_A(m, T/2)} \right. \\ & \left. + \sum_{k=1}^{T/2-1} 2\sqrt{\mathbf{S}_A(m, k)} \cos \left(\frac{2\pi kt}{T} + \phi_{m,k} \right) \right), \end{aligned} \quad (12)$$

where $\phi_{m,k}$ are independent random phases drawn uniformly from $[0, 2\pi)$. Finally, a full-column-rank mixing matrix $\mathbf{U} \in \mathbb{R}^{N \times M}$ is randomly generated and the correlated signal samples are obtained using (1), producing N correlated signals.

After generating signals, we apply our proposed multi-band sampling scheme, detailed in Sec. III-B, to sample these signals at the theoretical density $\frac{B}{N}$ and produce the reconstruction $\hat{\mathbf{X}}$.

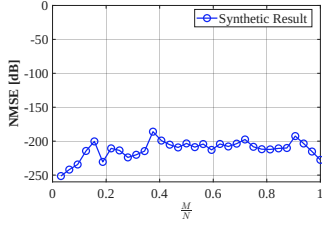


Fig. 4: Reconstruction NMSE across different $\frac{M}{N}$ ratio.

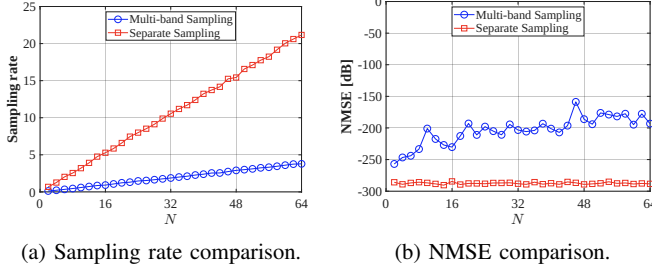


Fig. 5: The sampling rate and NMSE by different sampling schemes.

The Normalized Mean Squared Error (NMSE) is adopted to measure the reconstruction quality, defined as: $\text{NMSE} = \mathbb{E} \left[\left| \mathbf{X} - \hat{\mathbf{X}} \right|^2 \right] / \mathbb{E} \left[\left| \mathbf{X} \right|^2 \right]$.

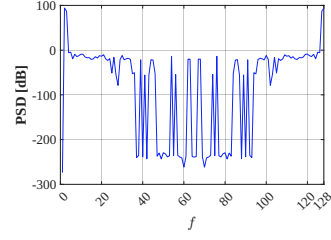
First, we evaluate the performance of multi-band sampling scheme under varying degrees of signal correlation, controlled by the ratio $\frac{M}{N}$. We fix the number of channels N and vary the ratio $\frac{M}{N}$. For each configuration, with $T = 512$ and 50 Monte Carlo trials, we compute the NMSE between the synthesized $\mathbf{X}$ and its reconstruction $\hat{\mathbf{X}}$. As shown in Fig. 4, the results demonstrate the robustness of our proposed scheme. It consistently maintains an NMSE below -180 dB at the theoretical sampling density $\frac{B}{N}$, regardless of the underlying correlation strength (from highly correlated, $\frac{M}{N} \rightarrow 0$, to weakly correlated, $\frac{M}{N} \rightarrow 1$).

Next, to quantify the sampling efficiency, we compare our multi-band scheme against a separate approach where each channel is sampled independently [5]. We fix the ratio $\frac{M}{N} = 0.5$ and set $T = 512$, with results averaged over 50 Monte Carlo trials. The results are summarized in Fig. 5. Fig. 5 (a) compares the sampling rates, which is defined as the total number of samples divided by T . It is evident that our proposed scheme requires a substantially lower sampling rate than the baseline approach. Crucially, Fig. 5 (b) confirms this efficiency gain does not compromise reconstruction fidelity. Despite the significant reduction in samples, the NMSE of our method remains at a comparably low level to the separate sampling scheme, confirming near-lossless performance. Taken together, these results validate that our method significantly reduces sampling costs while maintaining reconstruction accuracy.

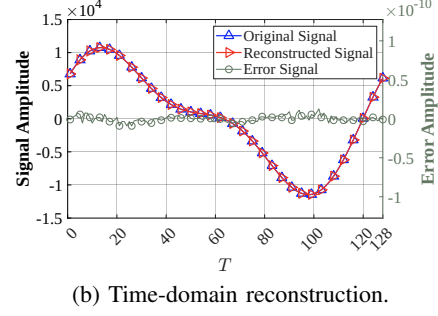
B. Results on real dataset

We test the multi-band sampling scheme on the public dataset Time Series Database Library (TSDL) [18], which contains 648 time series spanning domains such as finance, agriculture, meteorology, physics, production and sales.

We select a subset of series that satisfy the WSS assumption and truncate every series to a fixed length T for comparability. A collection of N series from different sources forms the raw correlated signals. We project the raw correlated signals into the latent uncorrelated component space via PCA to estimate the mixing



(a) Estimated PSD.



(b) Time-domain reconstruction.

Fig. 6: Real data reconstruction of a representative channel.

matrix and the unprocessed latent signals. For each estimated source we compute an empirical PSD.

Finite-sample effects and observational noise may induce spectral leakage and estimation noise. To mitigate spurious weak spectral components, we threshold each source's estimated PSD by setting values below 5% of that source's maximum PSD to zero. The thresholded spectra define the spectral supports used in subsequent processing and determine the total spectral bandwidth B . After preprocessing, we derive the thresholded latent signals $\mathbf{A}$ and corresponding correlated signals $\mathbf{X}$. Similar to the synthetic data construction, we then apply our multi-band sampling scheme at the resulting theoretical density $\frac{B}{N}$ to reconstruct the correlated signals.

Set $T = 128$ and $N = 10$, the results for a representative channel of multiple correlated signals $\mathbf{X}$ are presented in Fig. 6. Sec. IV-B displays the estimated PSD of this single channel. As illustrated in Sec. IV-B, the reconstructed signal closely tracks the original signal waveform. Visually, the two waveforms are nearly indistinguishable, and the error signal plotted below remains close to zero with a significantly smaller amplitude relative to the signal itself, providing a clear validation of our scheme's high-fidelity performance.

Quantitatively, the reconstruction achieves a total NMSE of -218.6915 dB between the observed signals $\mathbf{X}$ and the complete reconstruction $\hat{\mathbf{X}}$, confirming the effectiveness of our proposed method on real-world data.

V. CONCLUSION

In this paper, based on a latent source model, we establish a sampling theory for multiple correlated stochastic signals. We derive a sufficient lower bound on the total sampling density required for reconstruction with zero MSE, proving it equals the total spectral bandwidth of the latent sources divided by the number of correlated signals. Furthermore, we propose a constructive multi-band sampling scheme to achieve this bound via spectral partitioning. Experiments on synthetic data and real datasets confirm that our proposed scheme achieves near-lossless reconstruction at the derived sampling density, establishing a direct link between inter-channel correlation and sampling efficiency.

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